Ari Gestetner

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Education

Monash University | *Melbourne, Victoria, Australia*

Bachelor Science in Mathematics and Computational Science **2023-2025**

- Awarded with summer and winter vacation research scholarships
- First place in an IMC algorithmic trading competition for Monash students hosted by Monash Association of Coding (MAC)

Experience

Teaching Associate (Intro to Programming With Python & Intro to Computer Science) **Feb 2024 - Present**

Monash University

- Achieved 89% student satisfaction rating exceeding the faculty average of 82% by providing engaging, individualised and interactive classes and explanations
- Collaborated with fellow TAs to deliver applied classes, sharing feedback to drive continuous teaching improvements and enhance student learning outcomes

Research Scholarship — Dr. Buser Say **Jul 2023 - Feb 2025**

Monash University *Python, SCIP, Cplex, Gurobi, Seaborn, MILP, Constraint Programming*

- Benchmarked and presented different solvers and formulations finding results of over 2x decrease in solve time
- Investigated various linear programming constraint formulations based on neural network transition functions
- Refactored and extended SCIPPlan an automated planner for continuous time sequential planning problems over non-linear domains
- Assessed the performance of SCIPPlan in identifying optimal plans by formulating diverse problems with non-linear constraints
- Successfully derived feasible optimal solutions for a novel problem within a reasonable time frame
- Developed SCIPPlan+, an automated planner for discrete and continuous time planning problems with stochastic domains
- Researched the applications and performance of SCIPPlan+ over a variety of domains
- Introduced Github workflow to provide automated packaging and deployment to PyPi enabling a fast and easy development process
- Continued extending SCIPPlan to allow users to provide a system of odes to be solved and used in the planning process supporting engineers and physicists in their work
- Incorporated function interpolation, introducing flexibility to the planning process and allowing for more complex problems to be solved

Projects

Personal Website & Open Source Contribution *TypeScript, Nuxt, Vue, Vercel*

- Developed a personal portfolio site using Nuxt, Vue and Obsidian for content editing, managing the full deployment lifecycle via Vercel and Github actions
- Contributed to the Nuxt MDC open source ecosystem to fix a bug relating to MathJax support, providing a fix to issues spread across multiple repositories

Retail Sales Forecasting *R, Fable, Tidyverse*

- Analysed multi-decade Queensland retail turnover data to identify long term growth trends and predictable seasonal shopping patterns.
- Evaluated multiple statistical models to forecast future sales, specifically measuring the impact of major market disruptions like the 2020 pandemic and the speed of subsequent recoveries.

Finance API *Python, FastAPI, BeautifulSoup, PyTest*

- Architected a high-speed tool for stock data retrieval, using multithreading to scrape financial statements from multiple sources simultaneously for a 5x increase in speed.
- Engineered a Google Sheets integration used to manage and track $15,000 in real-world investments, ensuring system stability through automated unit testing.

Home Server & Infrastructure *Docker, Reverse Proxies, Linux, Networking*

- Manage a dedicated home server for self-hosting applications, utilising Docker deployment and container orchestration
- Configured vpn, reverse proxies and environments to enable secure and easy access and maintain high availability for hosted containers

Technical Skills

Languages: Python, R, Typescript, C, C++17, SQL, HTML/CSS

Frameworks: Flask, FastAPI, Django, PyTest, React, Vue, Nuxt, JUCE

Developer Tools: Git, Docker, RStudio, Linux, Vercel, GCP

Libraries: BeautifulSoup, Pandas, Matplotlib, Numpy/Scipy, Pytorch, Tidyverse, Fable, Cplex, Gurobi, SCIP

Academic Concepts: Analysis, Linear Algebra, Probability, Optimisation and OR, Graph Theory, Forecasting, Statistical Modelling, Econometric Theory